

JEE-Main-22-01-2025 (Memory Based)

[MORNING SHIFT]

Physics

$$\frac{B}{\mu_0}$$

Question: Find the dimensions of-

Options:

- (a) [AL]
- (b) [AL⁻¹]
- (c) [MAL]
- (d) [MALT⁻¹]

Answer: (b)

Question: Solid sphere of mass M, radius R exerts force F on a point mass. Now a concentric spherical mass M/7 is removed. What is new force?

Options:

- (a) F/7
- (b) 6/7 F
- (c) 5F/7
- (d) 3F/7

Answer: (b)

Question: Two drops of radii 2 cm and 8 cm are in contact. The radius of common surface is

Options:

- (a) 8/3 cm
- (b) 2 cm
- (c) 8 cm
- (d) 5/3 cm

Answer: (b)

Solution :

$$\frac{1}{r} = \frac{1}{2} + \frac{1}{8}$$

$$\frac{1}{r} = \frac{3}{8}$$

$$r = \frac{8}{3}$$

Question: The 7th harmonic of a closed organ pipe has same frequency as that of 4th harmonic of an open pipe. If two different gases with same bulk modulus with ratio of density $\frac{1}{3}$ and length of closed pipe is 10 cm. Find the length of open pipe

Options:

(a) $\frac{80}{7\sqrt{3}}$ cm

(b) $\frac{20}{7\sqrt{3}}$ cm

(c) $\frac{40}{7\sqrt{3}}$ cm

(d) $\frac{10}{7\sqrt{3}}$ cm

Answer: (b)

Solution :

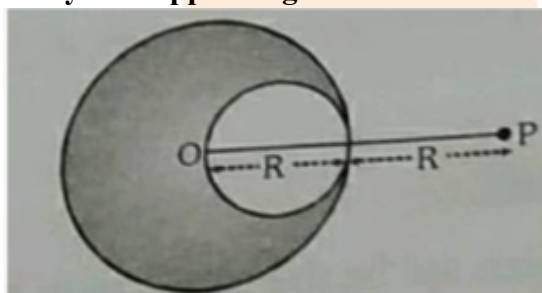
$$\frac{3.5}{2l_c} \sqrt{\frac{B}{p_c}} = \frac{4}{2l_0} = \sqrt{\frac{B}{p_0}} \quad \frac{p_c}{p_0} = \frac{1}{3}$$

$$l_0 = \frac{40}{3.5} \sqrt{\frac{p_c}{p_0}}$$

$$= \frac{80}{7} \times \sqrt{\frac{1}{3}}$$

$$l_0 = \frac{80}{7\sqrt{3}}$$

Question: A solid sphere of uniform density and radius R exerts a gravitational force of attraction F_1 on the particle P , distant $2R$ from the centre of the sphere. A spherical cavity of radius $R/3$ is now formed in the sphere as shown in figure. The sphere with cavity now applies a gravitational force F_2 on the same particle P . Find the ratio F_2/F_1 .



Options:

(a) $7/9$

(b) $9/7$

(c) $11/12$

(d) $12/11$

Answer: (c)

Question: A wire of length $a/2$ and charge density λ is kept along one of a cube of side length a . The wire's mid point is at the midpoint of the edge. Find the flux through the whole cube

Options:

(a) $\frac{\lambda a}{4\epsilon_0}$

(b) $\frac{\lambda a}{8\epsilon_0}$

(c) $\frac{\lambda a}{16\epsilon_0}$

(d) $\frac{\lambda a}{32\epsilon_0}$

Answer: (b)

Question: If a ball is thrown at 60 m/s at an angle of 30°. The ratio of height travelled in 1st second to the height travelled in the last second before reaching the maximum height is?

Options:

(a) 1:5

(b) 5:1

(c) 2:7

(d) 7:2

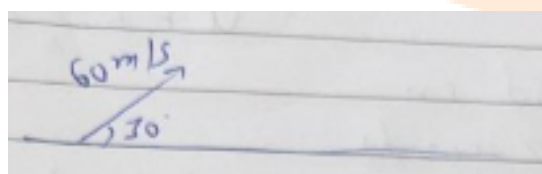
Answer: (b)

Solution :

$$\begin{aligned}\text{Height travelled in first 2 seconds} &= ut + \frac{1}{2}at^2 \\ &= 30(2) - \frac{1}{2} \times 10 \times (2)^2 \\ &= 60 - 20 = 40\end{aligned}$$

$$\begin{aligned}\text{Height travelled in last second before reaching more height} &= 45\text{ m} - 40\text{ m} \\ &= 5\text{ m}\end{aligned}$$

$$\frac{\text{Height in 1 second}}{\text{Height in last second}} = \frac{25}{5} = \frac{5}{1}$$



$$\begin{aligned}
 \text{Height in 1 seconds} &= ut + \frac{1}{2}at^2 \\
 &= 60 \sin 30(1) + \frac{1}{2} \times (-10) \times (1)^2 \\
 &= 60 \times \frac{1}{2} - \frac{1}{2} \times 10 \\
 &= 30 - 5 = 25 \text{ m}
 \end{aligned}$$

Time taken for reaching max height

$$\begin{aligned}
 &= \frac{4 \sin \theta}{g} \\
 &= \frac{60 \times \frac{1}{2}}{10} = 3
 \end{aligned}$$

$$\begin{aligned}
 \text{Max Height} &= ut + \frac{1}{2}at^2 \\
 &= 30(3) + \frac{1}{2} \times (-10) \times (-9) \\
 &= 90 - 45 = 45 \text{ m}
 \end{aligned}$$

Question: When a YDSE set up is immersed in a denser medium, then

(A) Assertion : Fringe width will decrease

(R) Reason : Speed of the wave will decrease but frequency remains constant

Options:

(a) Both (A) and (R) are correct and (R) explains (A)

(b) Both (A) and (R) are correct but (R) does not explain (A)

(c) (A) is correct but (R) is wrong

(d) (A) is wrong but (R) is correct

Answer: (a)

Question: If B represents magnetic field and μ represents permeability, then dimension

of $\frac{B}{\mu}$ is same as that of

Options:

(a) Length per unit current

(b) Current per unit length

(c) Length per unit charge

(d) Charge per unit length

Answer: (b)

Question: Moment of inertia of uniform disc of radius R and mass M about an axis passing through its centre and perpendicular to plane is I_1 . If a circular hole of diameter R whose rim passes through the centre is cut. The moment of inertia of the remaining part of the disc about a perpendicular axis, passing through the centre is I_2 . Find ratio of I_1 and I_2

Options:

- (a) 16/13
- (b) 32/13
- (c) 13/31
- (d) 13/32

Answer: (a)

Question: Two metals A and B having work function as 2.2 eV and 1.9 eV respectively are illuminated by monochromatic light of wavelength 550 nm. Which metal will show Photoelectric effect ?

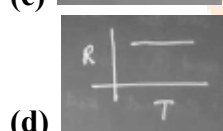
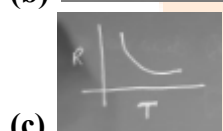
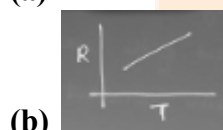
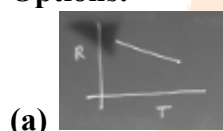
Options:

- (a) Only Metal A
- (b) Only Metal B
- (c) Both
- (d) None

Answer: (c)

Question: The correct graph of resistance vs temperature of a standard conducting wire will look like :

Options:



Answer: (b)

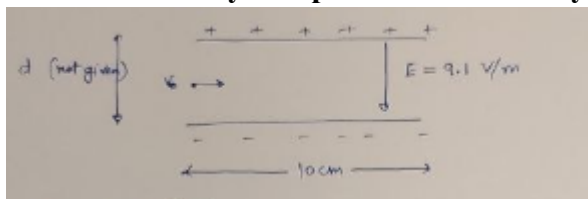
Question: Ice at -10°C is to be converted into steam at 110°C . Mass of ice is 10^{-1} kg. What amount of heat is required?

Options:

- (a) $\Delta Q = 730$ cal
- (b) $\Delta Q = 900$ cal
- (c) $\Delta Q = 1210$ cal
- (d) $\Delta Q = 870$ cal

Answer: (a)

Question: An electron is projected at $V_0 = 10^6$ m/s parallel to the plates of capacitor as shown. Find the y-component of velocity of electron as it comes out of plates.



Options:

- (a) 1.6×10^4 m/s
- (b) 1.6×10^6 m/s
- (c) 1.6×10^5 m/s
- (d) 1.6×10^3 m/s

Answer: (c)

Question: If two spherical black bodies of radii 0.2m and 0.8m which are at maintained at 400K and 800K respectively. Find ratio of power radiated by bodies.

Options:

- (a) 1/16
- (b) 1/256
- (c) 1/128
- (d) 1/144

Answer: (b)

Question: A capacitor is charged by battery to charge Q_1 . Now the battery is disconnected and dielectric slab of dielectric constant K is inserted between the gaps of the plates. Now charge on capacitor is Q_2 . Find Q_1/Q_2

Options:

- (a) 1
- (b) 1/2
- (c) 2
- (d) 2/3

Answer: (a)

Question: Find out equivalent capacitance for the situation show in figure.

Options:

(a)
$$C_{eq} = \frac{A\epsilon_0}{d} \left(\frac{K_1K_2 + K_2K_3 + K_3K_1}{K_1 + K_2} \right)$$

(b)
$$C_{eq} = \frac{A\epsilon_0}{d} \left(\frac{2K_1K_2 + K_2K_3 + K_3K_1}{2(K_1 + K_2)} \right)$$

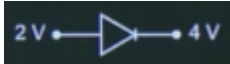
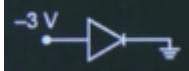
(c)
$$C_{eq} = \frac{A\epsilon_0}{d} \left(\frac{K_1K_2 + K_2K_3 + K_3K_1}{2(K_1 + K_2)} \right)$$

(d)
$$C_{eq} = \frac{A\epsilon_0}{2d} \left(\frac{K_1K_2 + K_2K_3 + K_3K_1}{(K_1 + K_2)} \right)$$

Answer: (b)

Question: From the given option, identify the diode connected in forward bias.

Options:

- (a) 
- (b) 
- (c) 
- (d) 

Answer: (b)

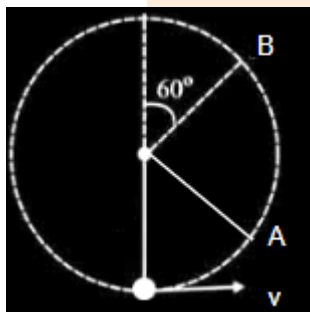
Question: Radius of electron in ground state for hydrogen is a_0 , then radius of electron in He^+ ion in 3rd excited state is a . Then a_0/a is

Options:

- (a) $1/2$
 (b) $1/4$
 (c) $1/16$
 (d) $1/8$

Answer: (d)

Question: The particle shown in figure is just able to complete the vertical circular motion. Find the ratio of kinetic energy at A to the kinetic energy at B.



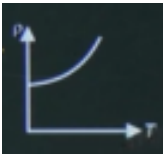
Options:

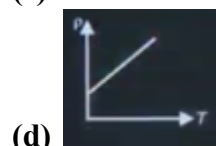
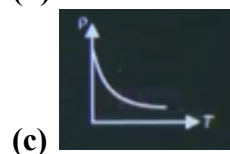
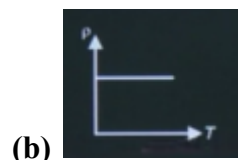
- (a)
 (b)
 (c)
 (d)

Answer: ()

Question: Which of the following graphs correctly represents the variation of resistivity (ρ) with temperature (T)?

Options:

- (a) 



Answer: (a)



JEE-Main-22-01-2025 (Memory Based)**[MORNING SHIFT]****Chemistry**

Question: What is the charge on metal and shape of complex of $[\text{NiCl}_4]^{2-}$ respectively?

Options:

- (a) +2, Tetrahedral
- (b) +2, Square planar
- (c) +4, Tetrahedral
- (d) +4, Square Planar

Answer: (a)

Question: Compare boiling point of given solutions

- (i) 10^{-4} NaCl
- (ii) 10^{-3} NaCl
- (iii) 10^{-2} NaCl
- (iv) 10^{-4} urea

Options:

- (a) I > II > III > IV
- (b) III > II > I > IV
- (c) II > I > III > IV
- (d) III > I > II > IV

Answer: (b)

Question: The correct decreasing order of electronegativity is

Options:

- (a) $\text{F} > \text{Cl} > \text{I} > \text{Br}$
- (b) $\text{Cl} > \text{F} > \text{Br} > \text{I}$
- (c) $\text{F} > \text{Cl} > \text{Br} > \text{I}$
- (d) $\text{Br} > \text{F} > \text{I} > \text{Cl}$

Answer: (c)

Question: Which of the following has maximum size out of Al^{3+} , Mg^{2+} , F^- , Na^+

Options:

- (a) Al^{3+}
- (b) Mg^{2+}
- (c) F^-
- (d) Na^+

Answer: (c)

Question: If the radius of the first orbit of the H atom is a_0 , then what is the radius of the first excited state of He^+ ion?

Options:

- (a) $\frac{a_0}{2}$
- (b) $2a_0$
- (c) a_0
- (d) $5a_0$

Answer: (b)

$$r_{He^+} = \frac{a_0 n^2}{Z} = \frac{a_0 \times 2^2}{2} = 2a_0$$

Question: Which has 7 electrons in the f subshell?

Options:

- (a) Eu^{3+}
- (b) Gd^{2+}
- (c) Eu^{2+}
- (d) Ce^{3+}

Answer: (c)

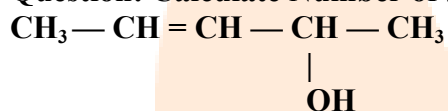
Question: The electrolysis product of which is $H_2S_2O_8$?

Options:

- (a) Dil H_2SO_4
- (b) $Cu SO_4(ag)$
- (c) Conc. H_2SO_4
- (d) None of these

Answer: (c)

Question: Calculate Number of stereoisomers of



Options:

- (a) 4
- (b) 2
- (c) 6
- (d) 8

Answer: (a)

Question: If $AlCl_3$ is electrolysed for 30 minutes using a current of 2A . How much of Al will be deposited at the cathode?

(F = 96500c, molar mass Al = 27g/Mol)

Options:

- (a) 0.1679
- (b) 0.2239
- (c) 0.3359
- (d) 0.4519

Answer: (c)

$$Q = I \times t = 2 \times 30 \times 60 = 3600 \text{ C.}$$

$$ne^- = \frac{Q}{F} = \frac{3600}{96500} \equiv 0.0373 \text{ mol } n_{Al} = \frac{0.0373}{3} = 0.0126 \text{ mol}$$

$$\text{wt of Al} = 0.0126 \times 27 = 0.335 \text{ g}$$

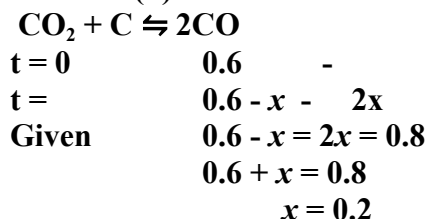
Question: $CO_2(g) + C(s) \rightleftharpoons 2CO(g)$

If initial pressure of CO_2 is 0.6 atm and after equilibrium is established, total pressure is 0.8 atm. Then, find K_p .

Options:

- (a) 0.8 atm
- (b) 0.4 atm
- (c) 0.5 atm
- (d) 0.2 atm

Answer: (b)



$$\therefore K_p = \frac{P_{\text{CO}_2}}{P_{\text{CO}}^2} = \frac{(0.4)^2}{(0.4)} = 0.4 \text{ atm}$$

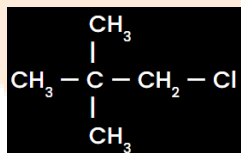
Question: In the Carius method of estimation of chlorine a compound of 180g produces 144g of AgCl. Find percentage composition of chlorine?

Options:

- (a) 20%
- (b) 36%
- (c) 23%
- (d) 25%

Answer: (a)

Question: Statement-1: $\text{CH}_3 - \text{O} - \text{CH}_2 - \text{Cl}$ will show nucleophilic substitution by $\text{S}_\text{N}1$ mechanism in protic medium



Statement-2: will not undergo nucleophilic substitution via $\text{S}_\text{N}2$ mechanism easily.

Options:

- (a) Statement-I and Statement-II both are correct
- (b) Statement-I and statement-II both are incorrect
- (c) Statement-I is correct but statement-II is incorrect
- (d) Statement-I is incorrect but statement-II is correct

Answer: (a)

Question: In CFSE (crystal field splitting energy), Δ_0 is zero for

Options:

- (a) $\text{K}_4[\text{FeC}(\text{N})_8]$
- (b) $\text{K}_3[\text{FeC}(\text{N})_6]$
- (c) $\text{K}_3[\text{FeF}_6]$
- (d) $\text{K}_2[\text{MnF}_6]$

Answer: (c)

$\text{n K}_3[\text{FeF}_6]$ the e^- configⁿ = $t_{2g}^3 e_g^2$

$$\text{CFSE} = 0.4 \times t_{2g}e^- + 0.6e_g e^-$$

$$= -0.4 \times 3 + 0.6 \times 2 = 0.$$

Question: Which of the following acids is present in a vitamin C ?

Options:

- (a) Ascorbic acid
- (b) Saccharic acid
- (c) Aspartic acid
- (d) Adipic acid

Answer: (a)

Question: Which of the following Electronegativity order is incorrect?

Options:

- (a) $\text{Mg} < \text{Be} < \text{B} < \text{N}$
- (b) $\text{Al} < \text{Si} < \text{C} < \text{N}$
- (c) $\text{S} < \text{Cl} < \text{O} < \text{F}$
- (d) $\text{Al} < \text{Mg} < \text{B} < \text{N}$

Answer: (d)

Question: An electron of He^+ is present in the 3rd excited state. Find its de-Broglie wavelength.

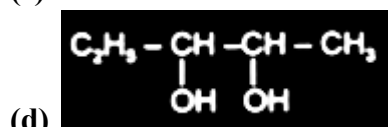
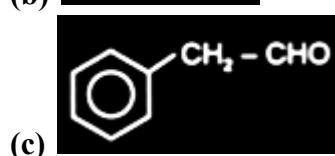
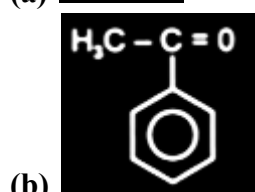
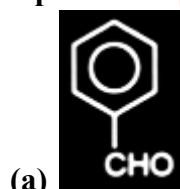
Options:

- (a) $6.28\text{\AA}$
- (b) $1.66\text{\AA}$
- (c) $3.32\text{\AA}$
- (d) $13.28\text{\AA}$

Answer: (a)

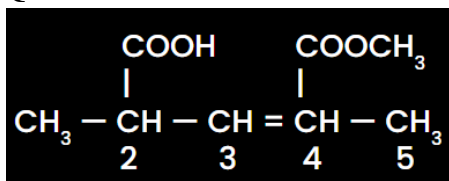
Question: Which will show a positive Fehling test?

Options:



Answer: (c)

Question: What is the IUPAC Name of the given compound?



Options:

- (a) 4-methoxy 2-methyl Pent-3-enoic acid carbonyl
- (b) 4-methoxy 3-methyl Pent-3-enoic acid carbonyl
- (c) 2-methoxy 4-methyl Pent-3-enoic acid carbonyl
- (d) 4-methoxy 2-methyl Pent-2-enoic acid carbonyl

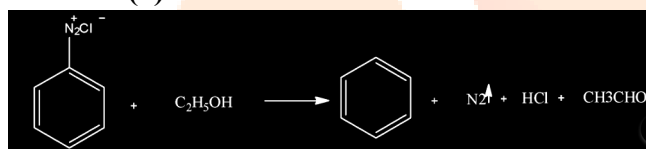
Answer: (a)

Question: When ethanol is treated with benzene diazonium chloride it forms:

Options:

- (a) Arenes
- (b) Methane
- (c) Amines
- (d) Ethyl alcohol

Answer: (a)



Question: If the work function of Cs and Fr is 1.9 & 2.7 eV. If light of $\lambda = 500 \text{ nm}$. Which element will show photoelectric effect?

Options:

- (a) Caesium
- (b) Fransium
- (c) Both have same
- (d) None of the above

Answer: (a)

$$E = \frac{12400}{\lambda(\text{\AA})} = \frac{12400}{5000} = 2.48 \text{ eV}$$

$$\text{KE} = h\nu - h\nu_0$$

$$h\nu > h\nu_0 \Rightarrow \text{Cs will show}$$

Question: Which of the following Statements is Incorrect?

Options:

- (a) Melting Point of cis-2-butene is greater than trans 2-butene
- (b) 2-methyl 2-butene has 2 Geometrical isomerism
- (c) Dipole moment of cis 2-butene is greater than trans 2-butene
- (d) In trans Isomer identical groups are opposite to each other

Answer: (b)

Question: 4f' configuration is possible for

- (a) Eu^{3+} , (b) Eu^{2+} , (c) Gd^{3+} , (d) Tb^{3+} , (e) Sm^{2+}

Options:

- (a) (a) and (c)
 (b) (b) and (c)
 (c) (d) and (e)
 (d) Only (c)

Answer: (b)

Atomic Number	Name	Symbol	Electronic configurations*		
			Ln	Ln ²⁺	Ln ³⁺
57	Lanthanum	La	5d ¹ 6s ²	5d ¹	4f ⁰
58	Cerium	Ce	4f ¹ 5d ¹ 6s ²	4f ²	4f ¹
59	Praseodymium	Pr	4f ³ 6s ²	4f ³	4f ²
60	Neodymium	Nd	4f ⁴ 6s ²	4f ⁴	4f ³
61	Promethium	Pm	4f ⁵ 6s ²	4f ⁵	4f ⁴
62	Samarium	Sm	4f ⁶ 6s ²	4f ⁶	4f ⁵
63	Europium	Eu	4f ⁷ 6s ²	4f ⁷	4f ⁶
64	Gadolinium	Gd	4f ⁷ 5d ¹ 6s ²	4f ⁷ 5d ¹	4f ⁷
65	Terbium	Tb	4f ⁹ 6s ²	4f ⁹	4f ⁸
66	Dysprosium	Dy	4f ¹⁰ 6s ²	4f ¹⁰	4f ⁹
67	Holmium	Ho	4f ¹¹ 6s ²	4f ¹¹	4f ¹⁰
68	Erbium	Er	4f ¹² 6s ²	4f ¹²	4f ¹¹
69	Thulium	Tm	4f ¹³ 6s ²	4f ¹³	4f ¹²
70	Ytterbium	Yb	4f ¹⁴ 6s ²	4f ¹⁴	4f ¹³
71	Lutetium	Lu	4f ¹⁴ 5d ¹ 6s ²	4f ¹⁴ 5d ¹	4f ¹⁴

Question: CO₂ gas is taken at 1 atm, 273K. Now it is allowed to pass through 0.1 M Ca(OH)₂ aq. Solution. Excess amount of Ca(OH)₂ is neutralized with 40 mL of 0.1 M HCl. Then find volume of Ca(OH)₂ initial taken if half of the amount of Ca(OH)₂ is reacted with CO₂

Options:

- (a) 40 mL
 (b) 20 mL
 (c) 80 mL
 (d) 50 mL

Answer: (a)

n_{HCl} : 4 millimoles

n_{Ca(OH)₂} = 2 millimoles

n_{Ca(OH)₂} = Reacting with 2 millimoles with CO₂

$$V = \frac{4}{0.1} = 40 \text{ ml}$$

Question: Match the column and choose the correct option

	Column-I(Properties)		Column-II (Order)
A	Electronegativity	1	B < C < N < O
B	Cationic size	2	Li > Mg > Be
C	Metallic Character	3	K > Mg > Al

D	Electron affinity	4	$\text{Cl} < \text{F} < \text{Br} < \text{I}$
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Options:

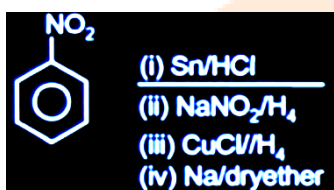
- (a) A-1, B-2, C-3, D-4
- (b) A-4, B-3, C-2, D-1
- (c) A-2, B-3, C-4, D-1
- (d) A-3, B-2, C-4, D-1

Answer: (a)

Question: How many compounds have the linear shape OF_2 , SO_2 , BeCl_2 , N_3^- , I_3^- , NO_2^+ , NO_2^- ?

Options:

Answer: (4)



Question:

Options:

Answer: (154)

Find molecular weight of A

JEE-Main-22-01-2025 (Memory Based)
[MORNING SHIFT]
Maths

Question: $f(x) = 7 \tan^8 x + 7 \tan^6 x - 3 \tan^4 x - 3 \tan^2 x$ $I_1 = \int_0^{\frac{\pi}{4}} f(x) dx$ $I_2 = \int_0^{\frac{\pi}{4}} x f(x) dx$ Find

$7I_1 + 12I_2$

Options:

(a) 1

(b) 2

(c) 3

(d) 4

Answer: (a)

$$f(x) = 7 \tan^8 x + 7 \tan^6 x - 3 \tan^4 x - 3 \tan^2 x$$

$$= 7 \tan^6 x \cdot \sec^2 x - 3 \tan^2 x \cdot \sec^2 x$$

$$I_1 = \int_0^{\frac{\pi}{4}} f(x) dx$$

$$= \int_0^{\frac{\pi}{4}} (7 \tan^6 x - 3 \tan^2 x) \sec^2 x dx$$

$$= \int_0^1 (7t^6 - 3t^2) dt$$

$$= t^7 - t^3 \Big|_0^1$$

$$= 0$$

$$\begin{aligned}
 I_2 &= \int_0^{\frac{\pi}{4}} x f(x) dx \\
 &= x \left[\tan^7 x - \tan^3 x \right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \tan^7 x + \tan^3 x \\
 &= 0 - \int_0^{\frac{\pi}{4}} \tan^3 x (\tan^2 x - 1) \tan^3 x \\
 &= - \int_0^{\frac{\pi}{4}} t^3 (t^2 - 1) dt \\
 &= - \int_0^{\frac{\pi}{4}} (t^5 - t^3) \\
 &= - \left[\frac{t^6}{6} - \frac{t^4}{4} \right]_0^{\frac{\pi}{4}} \\
 &= \frac{1}{4} - \frac{1}{6} = \frac{1}{12}
 \end{aligned}$$

Hence, $7I_1 + 12I_2 = 1$

Question: The number of 5 letter words which can be formed in alphabetical order such that M is always at middle taking every alphabet

Options:

- (a) 5143
- (b) 5148
- (c) 5144
- (d) 5149

Answer: (b)

M is the middle

$${}^{12}C_2 \times 1 \times {}^{13}C_2 = \frac{12 \times 11}{2} \times 1 \times \frac{13 \times 12}{2}$$

$$= 6 \times 11 \times 6 \times 13$$

$$= 5148$$

Question: $e^{5(\log_e x)^2 + 3} = x^8$ find the product of solutions

Options:

- (a) $e^{2/5}$
- (b) $e^{3/5}$
- (c) $e^{8/5}$
- (d) $e^{1/5}$

Answer (c)

$$e^5 (\log_e x)^2 + 3 = x^8$$

$$\Rightarrow 5(\log_e x)^2 + 3 = 8 \ln_e x$$

$$\Rightarrow 5t^2 + 3 = 8t$$

$$\Rightarrow 5t^2 - 8t + 3 = 0$$

$$\Rightarrow 5t^2 - 5t - 3t + 3 = 0$$

$$\Rightarrow 5t(t-1) - 3(t-1) = 0$$

$$t = 1, t = \frac{3}{5}$$

$$\ln_e x = 21 \quad \ln x = \frac{3}{5}$$

$$x = e \quad x = e^{\frac{3}{5}}$$

$$p = e^l \cdot e^{\frac{3}{5}}$$

$$= e^{\frac{3}{5}+1} = e^{\frac{8}{5}}$$

Question: Let the triangle PQR be the image of the triangle with vertices (1, 3), (3, 1), (2, 4) in the line $x + 2y = 2$. If the centroid of ΔPQR is the point (α, β) then $15(\alpha - \beta)$ is equation

Options:

(a) 20

(b) 21

(c) 22

(d) 23

Answer: (c)

$$\frac{1+3+2}{3}, \frac{3+1+4}{3} \quad x+2y-2=0$$

$$= \left(2, \frac{8}{3}\right) \quad \frac{x-2}{1} = \frac{y-\frac{8}{3}}{2} = -2 \frac{\left(1+\frac{16}{3}-2\right)}{5} = \frac{32}{15}$$

$$x = 2 - \frac{32}{15}, \quad y = \frac{8}{3} - \frac{64}{15}$$

$$= -\frac{2}{15} \quad = \frac{-24}{15} = \frac{-8}{5} = \frac{-24}{15}$$

$$15\alpha = -2$$

$$15\beta = -24$$

$$15(\alpha - \beta) = 22$$

Question: Set $A = \{1, 2, \dots, 10\}$ $B = \left\{ \frac{m}{n} : m < n, m, n \in A \right\}$ $(m, n) = 1$ $n(B) = ?$

Options:

(a) 30

(b) 31

(c) 32

(d) 33

Answer: (b)

$$A = \{1, 2, 3, \dots, 10\}$$

$$B = \left\{ \frac{m}{n} : m < n, m, n \in A \right\}$$

$$\frac{1}{10}, \frac{3}{10}, \frac{7}{10}, \frac{9}{10}$$

$$\frac{1}{9}, \frac{2}{9}, \frac{4}{9}, \frac{5}{9}, \frac{7}{9}, \frac{8}{9}$$

$$\frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}$$

$$\frac{1}{7}, \frac{2}{7}, \frac{3}{7}, \frac{4}{7}, \frac{5}{7}, \frac{6}{7}$$

$$\frac{1}{6}, \frac{5}{6}$$

$$\frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}$$

$$\frac{1}{4}, \frac{3}{4}$$

$$\frac{1}{3}, \frac{2}{3}$$

$$\frac{1}{2}$$

Total = 31

Question: A coin is tossed 3 times x is no of head following by tails $64[(M) + (\sigma^2)]$

Options:

(a) 71

(b) 40

(c) 60

(d) 48

Answer: (d)

$$x = 0 \{HHH, TTT, HTT, HHT\}$$

$$x = 1 \left\{ \begin{matrix} THH, HTH, TTH \\ THT \end{matrix} \right\}$$

$$P(x = 0) = \frac{1}{2} = P(x = 1)$$

$$\mu = \frac{1}{2}$$

$$\sigma^2 = \sum p_i \left(x_i - \frac{1}{2} \right)^2$$

$$= \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{4}$$

$$64 \left(\frac{1}{2} + \frac{1}{4} \right) = 32 + 16 = 48$$

2T, 1H

TTH, THT, HTT

2H, 1T

HHT, HTH

THH

Question: Two balls are selected at random one by one without replacement from the bag containing 4 white and 6 black balls. If the probability that the first selected ball is black given that the second selected is also black, is m/n where $\gcd(m, n) = 1$, then $m + n = ?$

Options:

- (a) 11
- (b) 12
- (c) 13
- (d) 14

Answer: (d)

$$4W, 6B \quad P\left(\frac{\text{First Black}}{\text{Second Black}}\right) = \frac{P(BB)}{P(WB) + P(BB)} = \frac{\frac{6}{10} \cdot \frac{5}{9}}{\frac{4}{10} \cdot \frac{6}{9} + \frac{6}{10} \cdot \frac{5}{9}} = \frac{30}{54} = \frac{5}{9} = \frac{m}{n} \quad m + n = 14$$

Question: $\sum_{r=1}^n T_r = \frac{(2n-1)(2n+1)(2n+3)(2n+5)}{64}$, then $\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{1}{T_r} \right) =$

Options:

- (a) 1
- (b) 0
- (c) 2
- (d) 3

Answer: (c)

$$T_r = S_n - S_{n-1}$$

$$= \frac{(2r-1)(2r-2)(2r+3)(2r+4)}{64} - \frac{(2r-3)(2r-1)(2r+1)(2r+3)}{64}$$

$$= 64(2r-1)(2r+1)(2r+3)[2r+5-(2r-3)]$$

$$T_n = \frac{1}{\theta}(2r-1)(2r+1)(2r+3)$$

$$\lim_{n \rightarrow \infty} \sum_{r=-\infty}^n \frac{1}{\ln}$$

$$\lim_{n \rightarrow \infty} \sum_{r=-\infty}^n \frac{\theta}{(2r-1)(2r+1)(2r+3)} \frac{[(2r+3)-(2r-1)]}{4}$$

$$= 2 \lim_{n \rightarrow \infty} \sum_{r=-\infty}^n \left[\frac{1}{(2r-1)(2r+1)} - \frac{1}{(2r+1)(2r+3)} \right]$$

$$= 2 \left[1 - \frac{1}{\infty} \right] = 2$$

Question: Parabola of equation $y^2 = 2\sqrt{3}x$ and circle of equation $y^2 + (x - 2\sqrt{3})^2 = 12$. Find the area inside the circle but outside the parabola?

Options:

(a) $2\pi - 8$

(b) $\pi - 8$

(c) $3\pi - 8$

(d) $\pi - 4$

Answer: (c)

$$y^2 = 2\sqrt{3}x$$

$$y^2 + (x - 2\sqrt{3})^2 = 12$$

$$2\sqrt{3}x + (2 - 2\sqrt{3})^2 = 12$$

$$2\sqrt{3}x + x^2 + 12 - 4\sqrt{3}x$$

$$x^2 = 2\sqrt{3}x$$

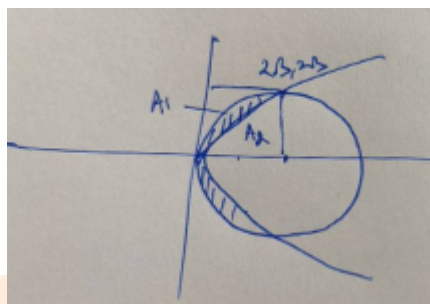
$$x = 0, 2B$$

$$y^2 = 2B \cdot 2B = 12$$

$$A_1 + A_2 = \frac{\pi \cdot 12}{4} = 3\pi$$

$$A_2 = \frac{2}{3} \cdot 2\beta \cdot 2\beta = 8$$

$$A_1 = 3\pi - 8$$



Question: $a_1, a_2, a_3, \dots$ are positive terms of GP if $a_1 a_5 = 28$ and $a_2 + a_5 = 29$. Then find $a_6 = ?$

Options:

(a) 780

(b) 782

(c) 784

(d) 786

Answer: (a)

$$a \cdot ar^4 = 28 \quad a_1 a_5 = 29$$

$$a_2 + a_4 = 25$$

$$a \cdot ar_4 = 28 \quad ar + ar_3 = 29$$

$$ar \cdot ar^3 = 28 \quad ar + \frac{28}{ar} = 29$$

$$(ar)^2 \cdot r^2 = 28 \quad (ar)^2 + 28 = 29r$$

$$ar = 28 \Rightarrow r^2 = \frac{1}{28} \quad (ar)^2 - 29(ar) + 29$$

$$ar = 1 \Rightarrow r^2 = 28 \quad (ar - 28)(ar - 1) = 0$$

$$ar=28 \quad ar=1$$

$$a_6 = ar_5 = (ar \cdot r^2 - r^2)$$

$$= 1 \times 28 \times 28$$

$$= 780$$

Question: $16[(\sec^{-1} x)^2 + (\operatorname{cosec}^{-1} x)^2]$ Find $m + M$ where m and M are the min and max values respectively .

Options:

(a) $20\pi^2$

(b) $22\pi^2$

(c) $2\pi^2$

(d) π^2

Answer: (a)

$$16 \left[\left(\sec^{-1} x \right)^2 + \left(\operatorname{cosec}^{-1} x \right)^2 \right]$$

$$16 \left(t^2 + \left(\frac{n}{2} - t \right)^2 \right), t = \sec^{-1} x \in [0, n] - \left\{ \frac{n}{2} \right\}$$

$$b(t) = 16 \left(2t^2 - \pi t + \frac{\pi^2}{4} \right)$$

$$= 32t^2 - 16\pi t + 4\pi^2$$

$$b'(t) = 64t - 16\pi = 0 \Rightarrow t = \frac{\pi}{4}$$

$$b(0) = 4\pi^2, b(\pi) = 20\pi^2$$

$$b\left(\frac{\pi}{4}\right) = 2\pi^2 - 4\pi^2 + 4\pi^2 = 2\pi^2$$

$$m = 2\pi^2 M = 20\pi^2$$

Question: If A be a 3×3 square matrix such that $\det(A) = -2$. If $\det(3\operatorname{adj}(-6\operatorname{adj}(3A))) = 2^n \times 3^m$, where $m \geq n$, then $4m + 2n$ is equal to

Options:

(a) 104

(b) 100

(c) 114

(d) 124

Answer: (a)

$$|A| = -2$$

$$|3\operatorname{adj}(-6\operatorname{adj}3A)| = 27|-6\operatorname{adj}3A|^2$$

$$= 27 \times (6^3)^2 |3A|^4$$

$$= 27 \times 6^6 \times (3^3)^4 \cdot |A|^4$$

$$= 27 \times 2^6 \times 3^6 \times 3^{12} \cdot 2^4 = 2^{10} \cdot 3^{21}$$

$$4m + 2n = 84 + 20 = 104.$$

Question: Let $f(x)$ be a real differentiable function such that $f(0) = 1$ and $f(x + y) = f(x)f'(y) + f(y)f'(x)$ for all $x, y \in \mathbb{R}$. Then

Options:

(a) 2525

(b) 1224

(c) 2500

(d) 1000

Answer: (a)

$$f(x+y) = f(x)f'(y) + f(y)f'(x)$$

$$y=0 \quad f'(x) + f'(0) \cdot f(x) = f(x)$$

$$\frac{dy}{y} = (1 - f'(0)) dx$$

$$\ln y = (1 - f'(0))x + c$$

$$0 = 0(1 - f'(0)) + c \rightarrow c = 0$$

$$\ln y = (1 - f'(0))x \rightarrow \frac{y^1}{y} = 1 - f'(0) \rightarrow f'(0) = \frac{1}{2}$$

$$\sum_{n=1}^{100} \ln f(x) = (1 - f'(0)) \frac{100(101)}{2}$$

$$= \frac{100 \times 101}{4} = 2525$$

Question: Hyperbola $\rightarrow$ F(1, 12) F1 (1,-14) passes through (1,6) Find Latus Rectum

Options:

(a) 20/7

(b) 30/7

(c) 40/7

(d) 50/7

Answer: (b)

$$Foci = F(1,12), F'(1,-14)$$

$$P(1,6)$$

$$PF = 6, PF' = 20 \Rightarrow PF' - PF = 14$$

$$\Rightarrow 2a = 14 \Rightarrow a = 7$$

$$\text{Also, } FF' = 16 \Rightarrow 2ae = 16 \Rightarrow e = \frac{8}{7}$$

$$b^2 = a^2(e^2 - 1) = 49\left(\frac{64}{49} - 1\right) = 15$$

$$LR = \frac{2b^2}{a} = \frac{30}{7}$$

Question: Number of equivalence relation on set $A = \{1, 2, 3\}$

Options:

(a) 5

(b) 8

(c) 2

(d) 0

Answer: (a)

Number of relations

$$\left\{ \underset{(1)}{(1,1)}, (2,2), (3,3) \right\} \left\{ \underset{(3)}{(1,1)}, (2,2), (3,3), (1,2), (2,1) \right\}$$

$$\left\{ \underset{(1)}{(1,1)}, (2,2), (3,3), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2) \right\}$$

$$= 5$$

Question: Z_1, Z_2, Z_3 lies on $|z| = 1$, arg are $\frac{\pi}{4}, -\frac{15}{4}, 0$.

$$\left| z \cdot \bar{z}_2 + z_2 \cdot \bar{z}_3 + z_3 \cdot \bar{z}_1 \right|^2 = (\alpha + \beta)\sqrt{2} \text{ Find } \alpha^2 + \beta^2.$$

Options:

(a) 29

(b) 2

(c) 4

(d) 25

Answer: (a)

$$z_1 = e^{i\pi/4}, z_2 = \bar{z}_1, z_3 = 1$$

$$(z, \bar{z}_2 + z_2 \bar{z}_3 + z_3 \bar{z}_1)^2 = (i + e^{-i\pi/4} + e^{-i\pi/4})^2$$

$$= [i + \sqrt{2} - \sqrt{2}i]^2 = 2 + (1 - \sqrt{2})^2$$

$$= 5 - 2\sqrt{2}$$

$$\alpha = 5\beta = -2$$

$$\alpha^2 + \beta^2 = 29$$

Question: $\frac{dx}{dy} + \frac{x}{y^2} = \frac{1}{y^2}$ then $x(0) = 1$ and $x\left(\frac{1}{2}\right) = ?$

Options:

(a) 1-e

(b) 2-e

(c) 3-e

(d) 4-e

Answer: (c)

$$\left. \begin{aligned} x.e^{\frac{1}{y}} &= \int \frac{2}{y^3} e^{\frac{1}{y}} dy \\ &= \int te' dt \quad \frac{-1}{y} = t \\ &= \int te' dt \\ &= e^t (1-t) + c \\ &= e^{\frac{1}{y}} \left(1 + \frac{1}{y} \right) + ce^{\frac{1}{y}} \end{aligned} \right\}$$

$$y = +1, x = 1 \quad 1 = 1 + 1 + c.e^1$$

$$c = -1/e$$

$$x\left(\frac{1}{2}\right) = 3 - \frac{1}{e} \cdot e^2$$

$$= 3 - e$$

Question: $S_2 : (2, 5)$ is centre and $r = r$, $r \in [\alpha, \beta]$ such that both cut at different point.

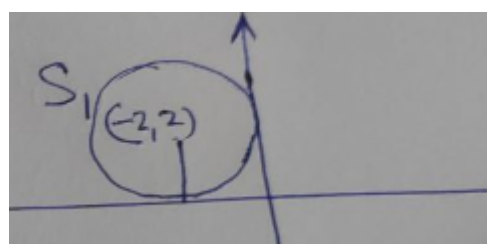
Options:

(a) $r \in [3, 7]$

(b) $r \in [2, 7]$

(c) $r \in [3, 5]$

(d) None



Answer: (a)

$$C_1 C_2 = \sqrt{(2+2)^2 + (5-2)^2} = 5$$

$$|r-2| < 5 < r+2$$

$r > 3$ and $r < 7$ or

$$r \in [3, 7]$$

Question: Shortest distance between $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-1}{4}$ and $\frac{x+2}{7} = \frac{y-2}{8} = \frac{z+1}{2}$

Options:

(a) $\frac{80}{\sqrt{1277}}$

(b) $\frac{88}{\sqrt{1277}}$

(c) $\frac{8}{\sqrt{1277}}$

(d) $\frac{87}{\sqrt{1277}}$

Answer: (b)

$$S.D \frac{|\begin{vmatrix} 3 & 0 & 2 & 2 & 3 & 4 & 7 & 8 & 2 \\ \hat{i} & \hat{j} & \hat{k} & 2 & 3 & 4 & 7 & 8 & 2 \end{vmatrix}|}{|-26\hat{i} + 24\hat{j} - 5\hat{k}|} = \frac{|-78-10|}{\sqrt{1277}} = \frac{88}{\sqrt{1277}}$$

Question: $\sum_{r=0}^5 \frac{{}^{11}C_{2r-1}}{2r+2} = ?$

Options:

(a) 2048/6

(b) 2048/12

(c) 2047/12

(d) 2047/6

Answer: (d)

$$\sum_{r=0}^5 \frac{{}^{11}C_{2r-1}}{2r+2} = \int_0^1 (1+x)^{11} + (1-x)^{11} dx$$

$$= \frac{(1+x)^{12}}{12} + \frac{(1-x)^{12}}{12} \Big|_0^1$$

$$= \frac{12^{12}}{12} - \frac{1}{12} + 0 - \frac{1}{12}$$

$$= \frac{12^{12} - 2}{12} = \frac{2047}{6}$$